PERCONA LIVE 2023

Extra Functionalities in PMM

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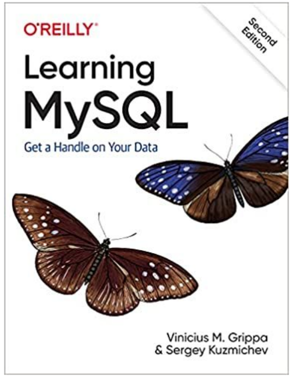
About me

Percona Database Engineer for 6 years

Working with databases for 18 years

Co-author of the book Learning MySQL



Agenda

- What to monitor in a database?
- Why Monitoring?
- Agent vs Agentless Monitoring
  - Local Agent Monitoring
  - Agentless Monitoring
- ProcFS Plugin
- Integrating ProcFS with PMM
- Alerting and Sending Notifications
- Questions

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What to monitor in a database?

What To Monitor In a Database?

- Response time
- QPS
- Workload Behavior
- Infrastructure components (network, firewalls, storage, power supply, etc…)
- Resource utilization, security breaches
- ….?

Why Monitoring?

Why Monitoring?

- It helps diagnosing past and ongoing performance issues.
- It helps to identify issues proactively.
- It helps finding optimization opportunities and cost savings.
- It helps you sleep better at night.

Why Monitoring?

Agent vs Agentless Monitoring

Agent Monitoring

- Pros:
  - Greater monitoring depth is possible.
  - Some agents provide a way to run custom scripts.
- Cons:
  - Agents require that you install them on your systems, so there would be an additional item to install, manage, and secure.
  - Since they require something be installed, it may be more difficult to gain approval from systems teams to roll out agents.

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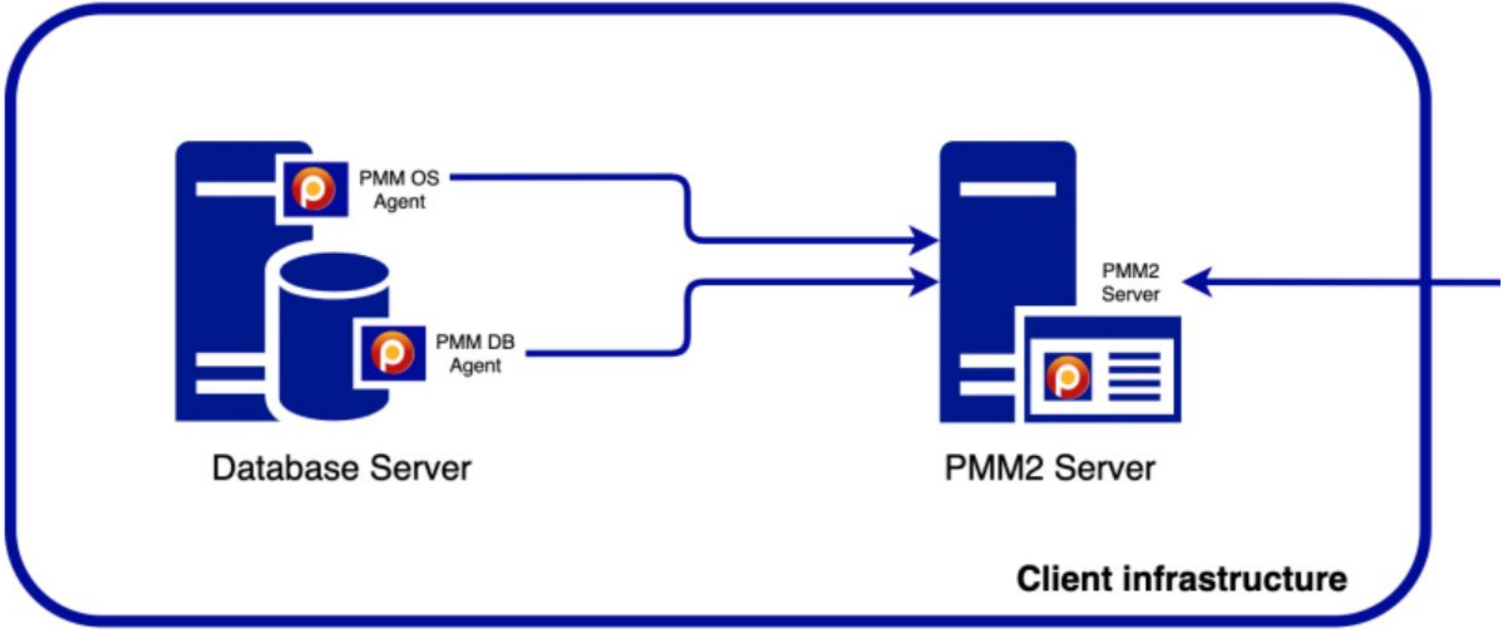
Local Agent Monitoring



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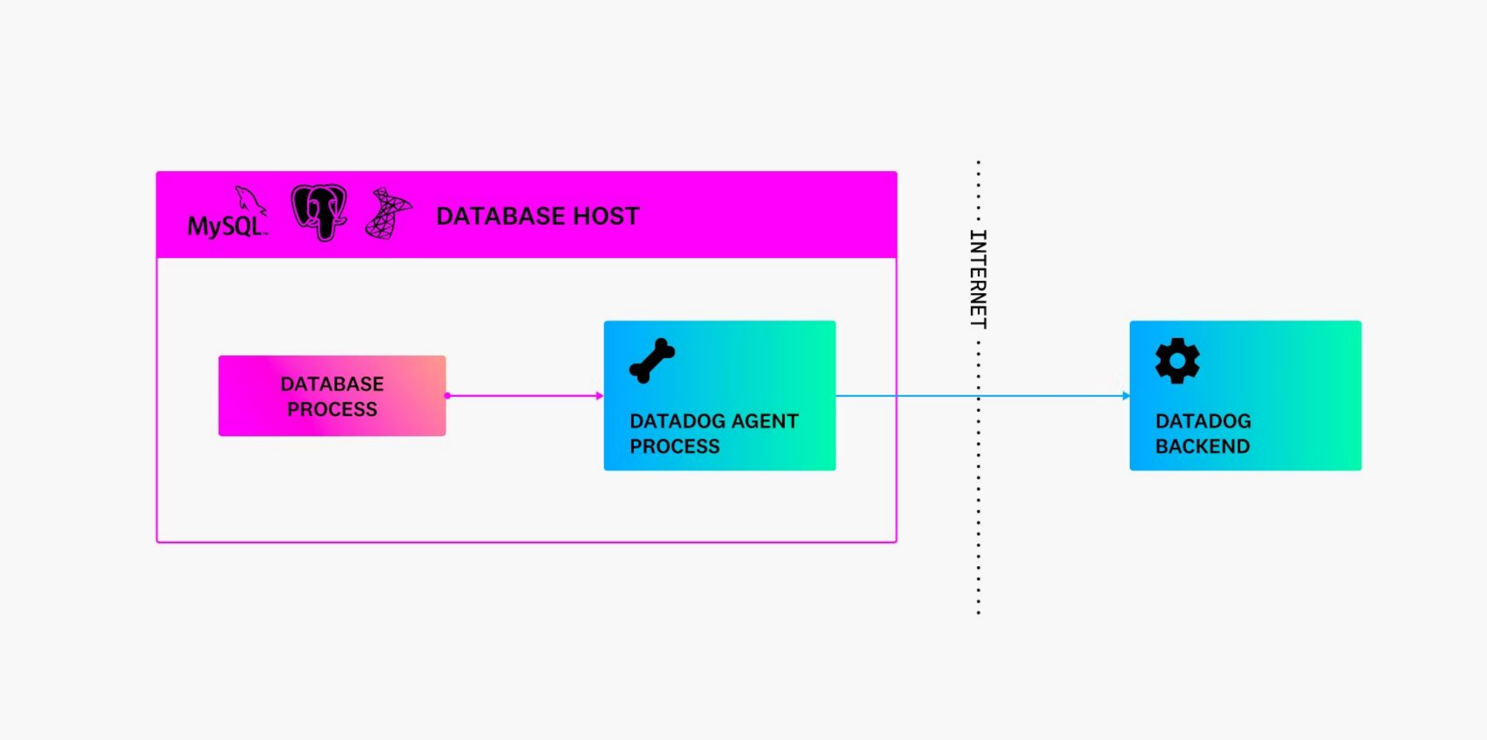
Local Agent Monitoring



Agentless Monitoring

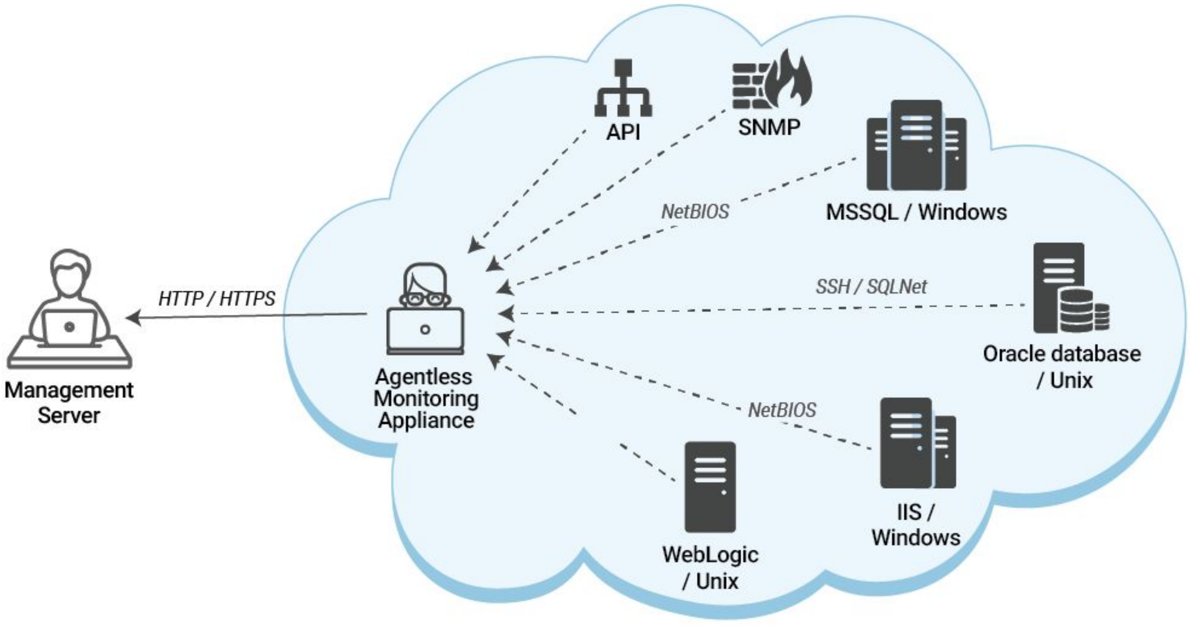
Pros:

- Agentless approaches can help reduce administrative overhead, since there is no separate agent to install or manage.
- Agentless methods may be easier to get approval to use from systems teams, since again no third-party utilities need be installed.

Cons:

- You will likely still need to take some action on the monitored host, for example, defining credentials, or loading plugins.
- The available checks is usually limited to whatever metrics are supported.
- Agentless methods can introduce higher load to the central monitoring server, so may not scale as well as agents.

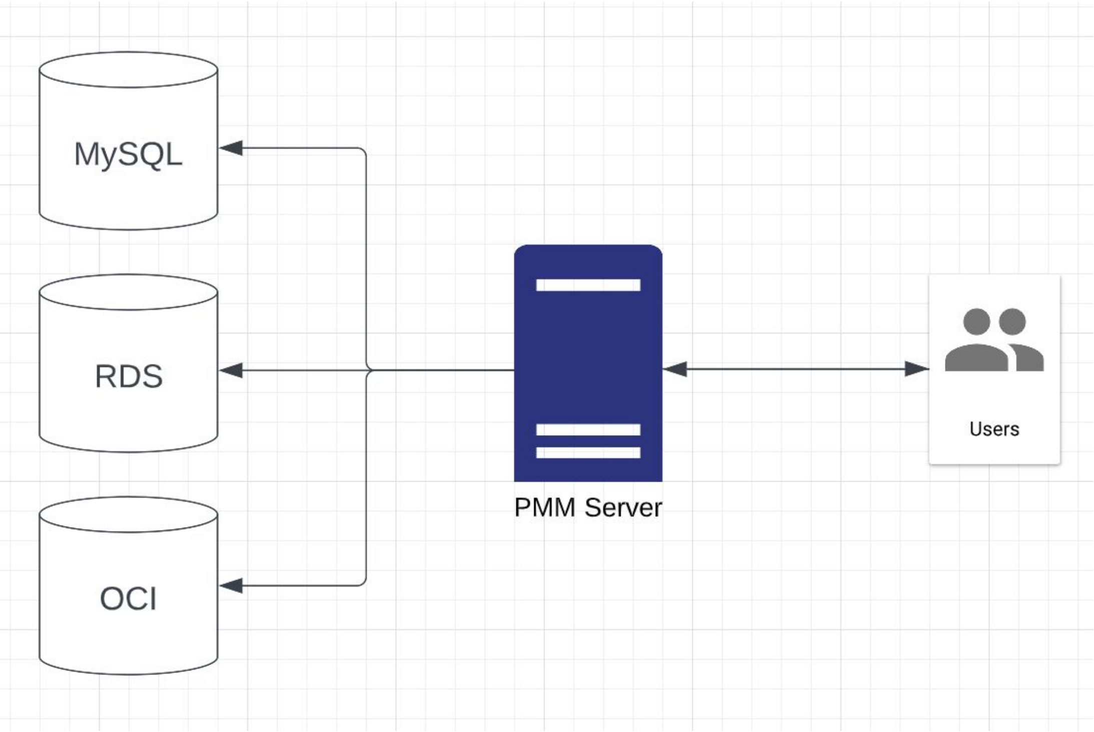
Agentless Monitoring



Agentless Monitoring of an IT infrastructure

Source: https://www.eginnovations.com/blog/agentless-vs-agent-based-monitoring/

Agentless Monitoring



ProcFS Plugin

ProcFS Plugin

ProcFS plugin provides access to the Linux performance data.

The plugin does the following:

- Reads selected files from the /proc file system and the /sys file system.
- Populates the file names and their content as rows in the INFORMATION_SCHEMA.PROCFS view.
- User has control of what is exposed with the `procfs_files_spec`parameter(static).

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ProcFS Plugin Caveats

At the moment, it only works with Percona Server 😢

```
18:27:48 UTC - mysqld got signal 11 ;

[...]

where mysqld died. If you see no messages after this, something went terribly
wrong...

stack_bottom = 7f128441fd80 thread_stack 0x100000

[...]

Trying to get some variables.

Some pointers may be invalid and cause the dump to abort.

Query (7f123c1d9b40): SELECT * FROM INFORMATION_SCHEMA.PROCFS WHERE FILE =
'/proc/version'

Connection ID (thread ID): 12

Status: NOT_KILLED
```

ProcFS Plugin Caveats

The plugin only works within the same Percona Server version

```
mysql> INSTALL PLUGIN procfs SONAME 'procfs.so';

ERROR 1126 (HY000): Can't open shared library 'procfs.so' (errno: 0 API version for

INFORMATION SCHEMA plugin is too different)
```

Installing ProcFS Plugin

```
mysql> INSTALL PLUGIN procfs SONAME 'procfs.so';

mysql> GRANT ACCESS_PROCFS ON *.* TO

'user'@'host';
```

Even root user needs this privilege!

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Using ProcFS Plugin

```
mysql> SHOW GLOBAL VARIABLES LIKE
'procfs_files_spec'\G
*************************** 1. row
***************************
Variable_name: procfs_files_spec
        Value:
/proc/cpuinfo;/proc/irq/*/*;/proc/loadavg;/pr
oc/net/dev;/proc/net/sockstat;/proc/net/sockst
at_rhe4;/proc/net/tcpstat;/proc/self/net/netst
at;/proc/self/stat;/proc/self/io;/proc/self/n
uma_maps/proc/softirqs;/proc/spl/kstat/zfs/arc
stats;/proc/stat;/proc/sys/fs/file-nr;/proc/ve
rsion;/proc/vmstat
1 row in set (0.00 sec)
```

Using ProcFS Plugin

```
mysql> SELECT * FROM INFORMATION_SCHEMA.PROCFS WHERE FILE =
'/proc/version'\G
*************************** 1. row ***************************
    FILE: /proc/version
CONTENTS: Linux version 5.15.0-1028-aws (buildd@lcy02-amd64-017)
(gcc (Ubuntu 11.3.0-1ubuntu1~22.04) 11.3.0, GNU ld (GNU Binutils
for Ubuntu) 2.38) #32-Ubuntu SMP Mon Jan 9 12:28:07 UTC 202
```

Integrating ProcFS with Prometheus

It is possible to integrate with Node Exporter

You can compile or use podman/docker

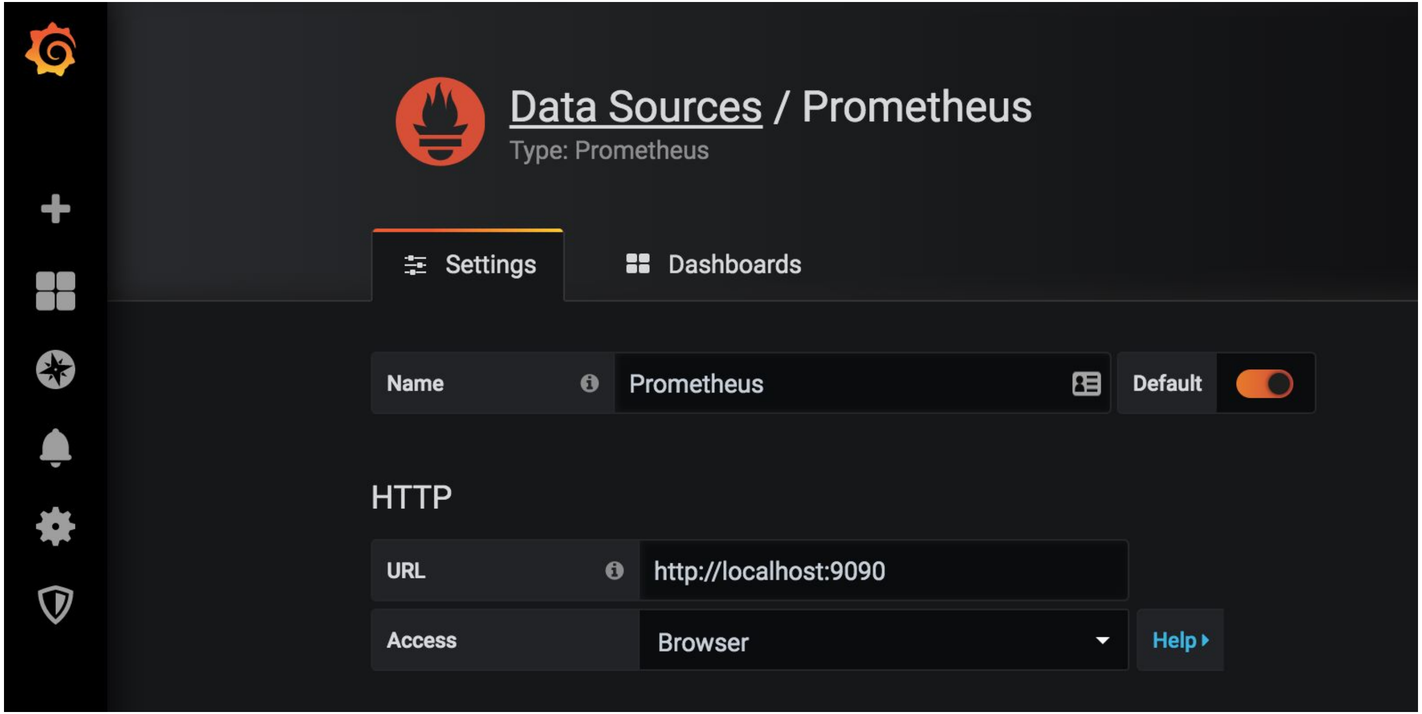
https://www.percona.com/blog/procfs-udf-different-approach-to-agentless-operating-system-observability-in-your-database/

Have `mysqlprocfs` collector

https://github.com/ihanick/node_exporter

It is possible to integrate with any monitoring tool. For example with Grafana.

Integrating ProcFS with Prometheus



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Integrating ProcFS with Prometheus

```
$ docker run -p 9100:9100 -d --name procfsdemo
--restart always
docker.io/perconalab/node_exporter:procfs
--collector.mysqlprocfs="vgrippa:fosdem2023@tcp(
18.209.66.223:3306)"
```

```
$ curl  127.0.0.1:9100/metrics
```


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Demo Integrating ProcFS with Grafana/PMM

Integrating ProcFS with PMM

Integrating ProcFS with PMM

Services Agents Nodes

Delete

	ID	Service Type	Service name	Node ID	Addresses	Port	Other Details
	/service_id/279a810b-bff6-4dbb-8a34-dd3a0856d9d1	MySQL	mysql_remote	/node_id/11bab135-99a1-4dd4-b699-3e191e9b448d	18.209.66.223	3306	
	/service_id/f8db1b87-ec64-422c-b2fe-b8d25b22d10e	PostgreSQL	pmm-server-postgresql	pmm-server	127.0.0.1	5432	database_name: postgres
	/service_id/f86195ea-1e78-4f34-aaa1-dc7d8201e900	External	procfsdemo	/node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28			group: external
	/service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d	External	procfstest-node	/node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28			group: external

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Integrating ProcFS with PMM

```
# pmm-admin inventory list nodes
Nodes list.

Node type     Node name          Address         Node ID
GENERIC_NODE  ip-172-30-3-118    172.30.3.118    /node_id/00bfdf1d-89bd-4e64-ad07-cee2d0b909a4
GENERIC_NODE  pmm-server         127.0.0.1       pmm-server
REMOTE_NODE   mysql_remote       18.209.66.223   /node_id/11bab135-99a1-4dd4-b699-3e191e9b448d
REMOTE_NODE   procfsdemo         172.17.0.3      /node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28
```


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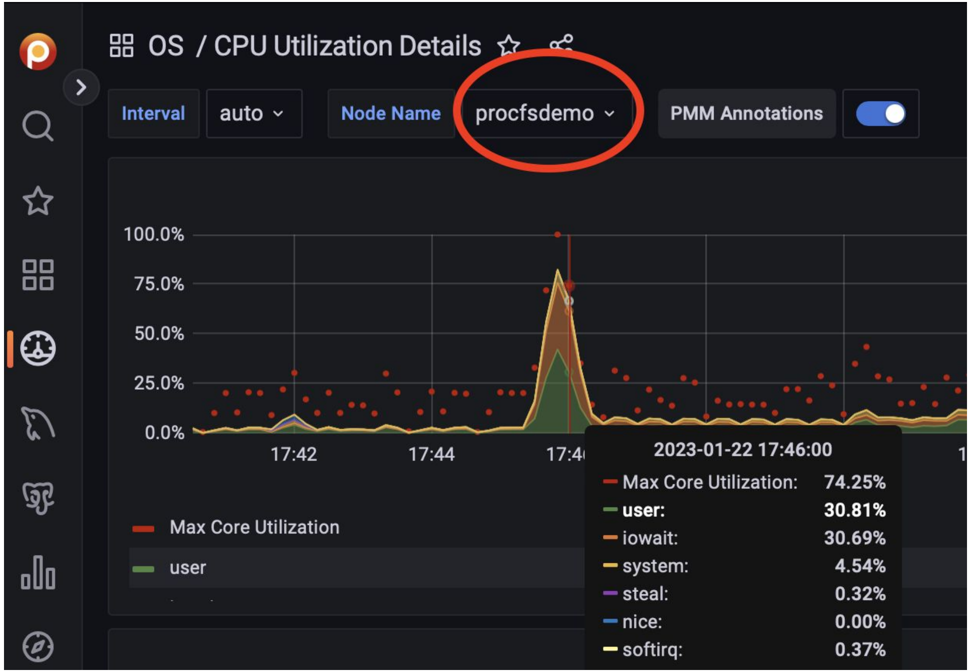
Integrating ProcFS with PMM

```
# pmm-admin inventory add service external --name=procfstest-node
--node-id=/node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28


External Service added.

Service ID      : /service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d

Service name    : procfstest-node

Node ID         : /node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28

Environment     :

Cluster name    :

Replication set:

Custom labels   : map[]

Group           : external
```

Integrating ProcFS with PMM

```
# pmm-admin inventory add agent external
--runs-on-node-id=/node_id/00bfdf1d-89bd-4e64-ad07-cee2d0b909a4
--service-id=/service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d --listen-port=9100


External Exporter added.
Agent ID                : /agent_id/91d6b7c8-61d6-4707-8b83-afaf25ac1db0
Runs on node ID         : /node_id/00bfdf1d-89bd-4e64-ad07-cee2d0b909a4
Service ID              : /service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d
Username                :
Scheme                  : http
Metrics path            : /metrics
Listen port             : 9100
Disabled                : false
Custom labels           : map[]
```

Integrating ProcFS with PMM

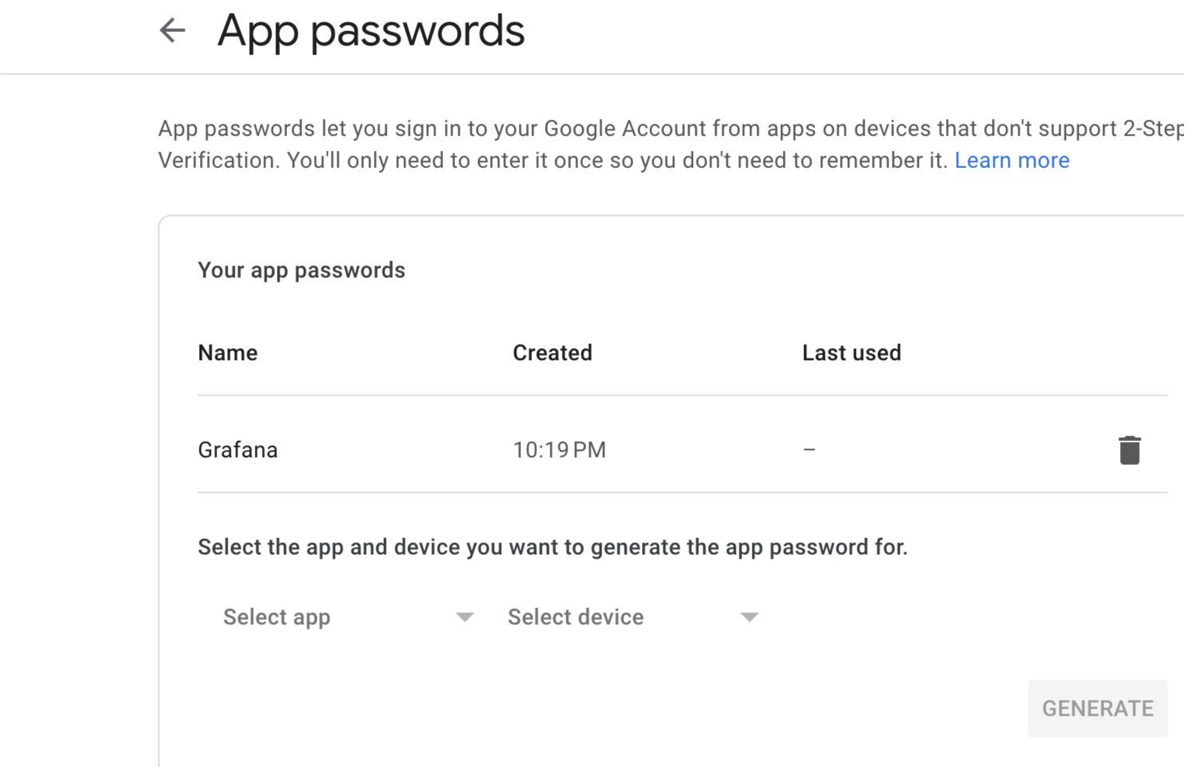


Alerting and Sending Notifications

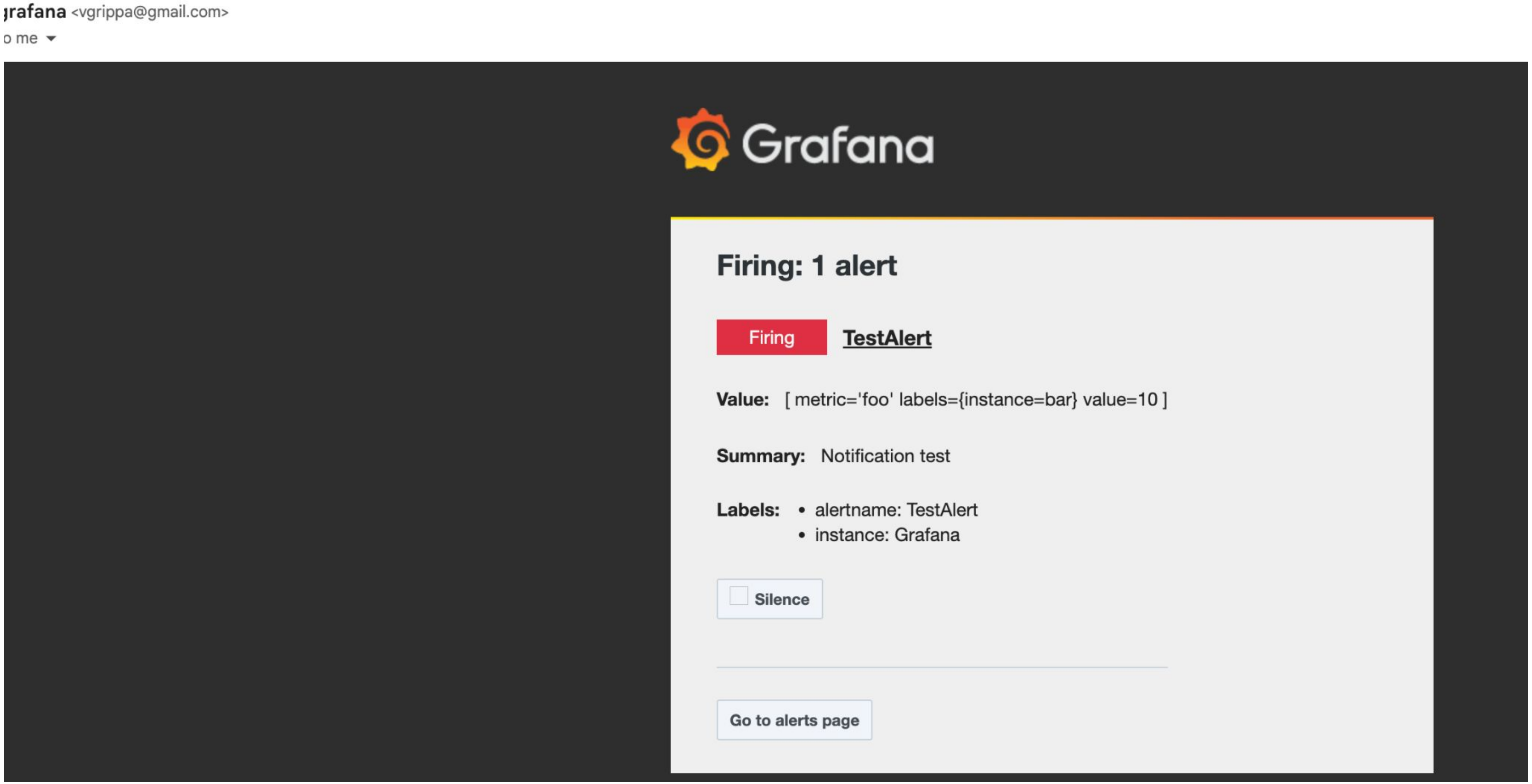
Sending Notifications - GMAIL

```
docker create --volume /srv \
--name pmm-data percona/pmm-server:2.36  /bin/true

docker run -d \
-p 80:80 \
-p 443:443 \
--volumes-from pmm-data \
--name pmm-server \
--restart always \
 -e GF_SMTP_ENABLED=true \
 -e GF_SMTP_HOST=smtp.gmail.com:587 \
 -e GF_SMTP_USER=vgrippa@gmail.com \
 -e GF_SMTP_PASSWORD="123456" \
 -e GF_SMTP_SKIP_VERIFY=false \
 -e GF_SMTP_FROM_NAME=grafana \
 -e GF_SMTP_FROM_ADDRESS=vgrippa@gmail.com \
percona/pmm-server:2.36
```

Sending Notifications - GMAIL



Sending Notifications - GMAIL



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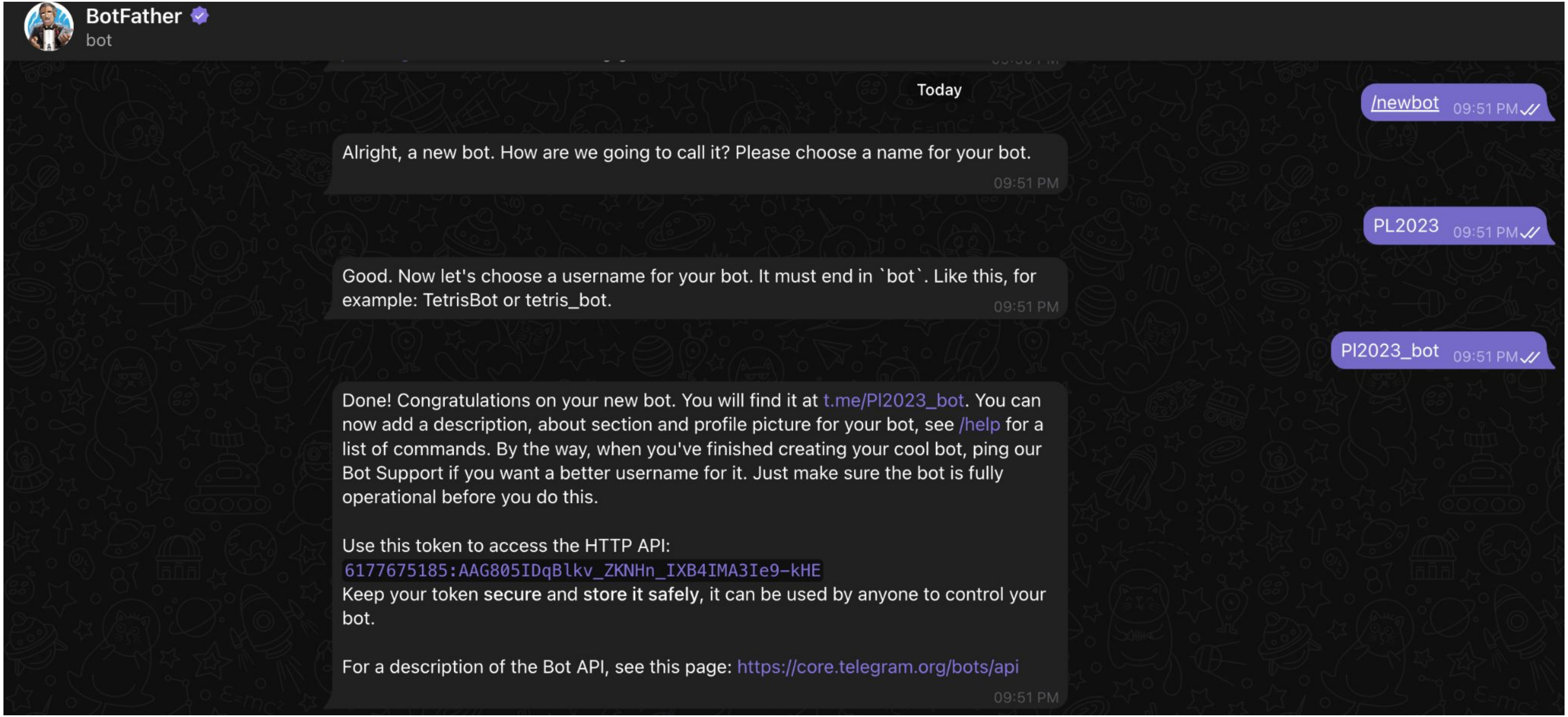
Sending Notifications - Telegram

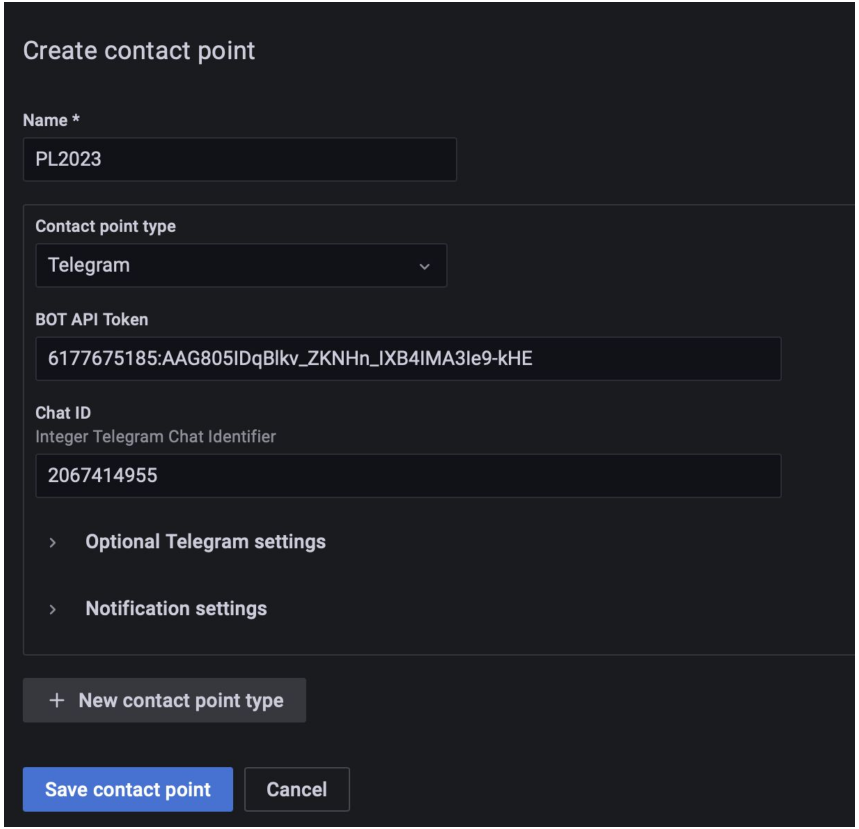


Sending Notifications - GMAIL

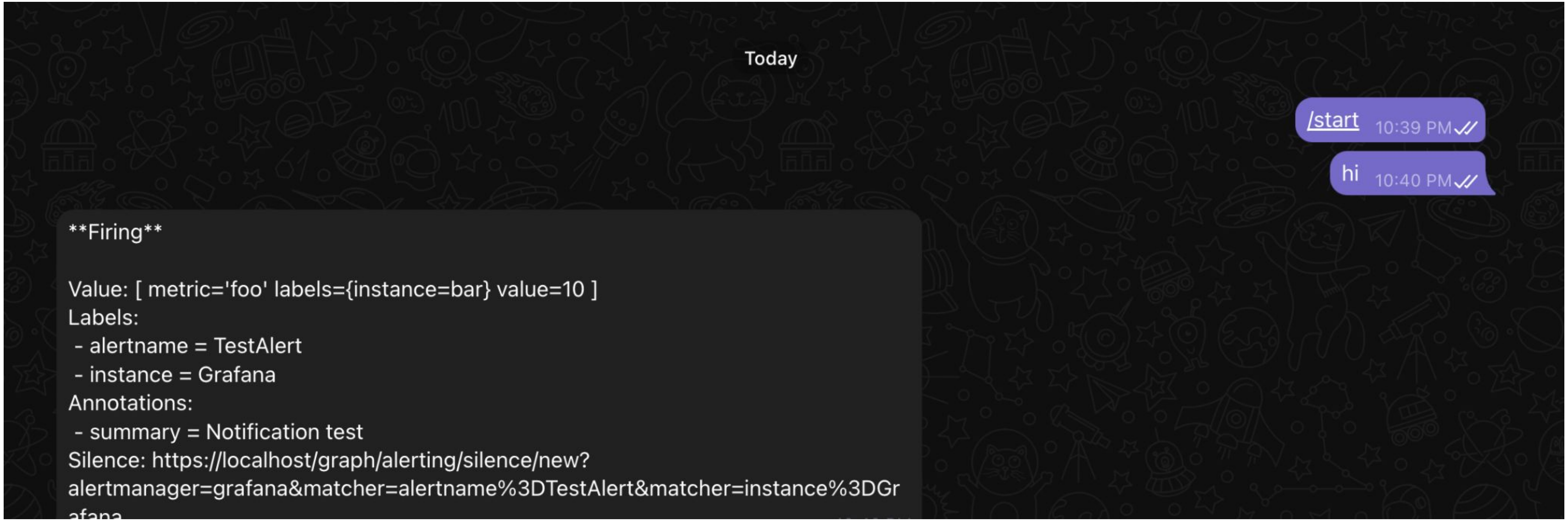
https://api.telegram.org/bot6177675185:AAG805IDqBlkv_ZKNHn_IXB4IMA3Ie9-kHE/getUpdates

"message":{"message_id":2,"from":{"id":2067414955,"is_bot":false,"first_name":"Vinicius","last_name":"Grippa","username":"vgrippa","language_code":"en"},"chat":{"id":2067414955,"first_name":"Vinicius","last_name":"Grippa","username":"vgrippa","type":"private"},"date":1684816839,"text":"hi"}}]

Sending Notifications - Telegram



Sending Notifications - Telegram



Demo?

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Questions?

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Thank you!